Critical Literature Review

This review connects research on intelligent tutoring systems, diagnosis, gamification and learning analytics to the design of MathVerity. The main evaluation criterion is independent performance on new items after practice.

Selection Method

This is a targeted critical review, not a reproducible systematic review or a PRISMA study. The search was done on 22 October 2026. Search topics included K-12 intelligent tutoring systems, mathematics formative assessment, knowledge tracing, gamification and parent learning analytics. Open research directly relevant to architecture or evaluation was included. Higher-education-only studies and unverifiable bibliographic records were not used as the core evidence base.

The review includes 12 scientific publications from 2024–2025 and two official documents, exceeding the minimum of 12 sources and 8 scientific publications. It is a selected body of evidence rather than an exhaustive account; several studies concern other age groups. Public full texts are linked through the listed pages.

Intelligent Assistance and Independent Performance

The ITS synthesis [S01] supports using a high-quality digital practice tool as the control condition. The presence of AI alone does not explain educational benefit. The review of mathematics ITS [S03] also suggests examining changes in learning activity rather than only automated answer checking. The design therefore centres on an attempt–assistance–independent transfer cycle rather than an unrestricted chat.

The experiment by Bastani and colleagues [S02] demonstrates the risk of relying on ready-made AI solutions. For MathVerity, this supports restricted hints and separate tests without AI. This is an engineering interpretation of research with high school students, not an established effect for the intended users.

Gamification

The gamification review [S08] and the mathematics-game randomised trials [S06] do not justify treating additional game elements as a guarantee of learning gains. The MVP awards XP for completing an independent attempt and achievements for evidenced progress. Public comparisons, penalties for missing a day and rewards for time spent are excluded. Learning outcomes and motivation are measured separately; a causal effect of gamification is not claimed without a separate comparison.

Analytics and Family Support

Research with grades 5–6 [S07] shows that digital traces capture only part of learning activity. The review of school analytics [S09] emphasises careful interpretation and data handling. MathVerity therefore reports time within the platform, the number of independent items, the date of the last assessment and an "insufficient evidence" status separately.

The study of teachers' practices [S10] supports access to the evidence behind a diagnosis. The review of intelligent dashboards [S11] supports evaluating actions taken after reading a report, rather than prediction accuracy alone. Applying these findings to parents is a UX hypothesis. Research with preschool families [S12] provides a further rationale for family information support but does not establish the effectiveness of a parent dashboard for grades 5–6.

Research Gap and Design Response

The publications support individual mechanisms but do not establish that their combination is effective for the chosen audience. MathVerity's engineering contribution is a coherent implementation of deterministic assessment, restricted hints, independent assessment and an explainable report. No claim of global novelty or absence of comparable systems is made. The primary criterion is independent performance on new items, rather than more clicks, XP or AI-assisted answers.

Scientific Publication Matrix

Table 1. Analysis of scientific publications

Code and year	Method	Finding	Limitation	Design response
S01 2025	Review of 28 studies involving 4,597 students	ITS outcomes are generally positive; advantages are smaller against other digital practice tools.	Heterogeneous interventions and durations; results do not prove the benefit of a particular LLM.	Compare MathVerity against an identical practice tool without AI, not only against no instruction.
S02 2025	Field experiment with high school students	Unrestricted AI improved practice performance but could reduce later independent performance; pedagogical restrictions mitigated the risk.	High school students in one setting; not direct evidence for grades 5–6.	Separate assisted practice from independent assessment. Author-affiliation correction considered: DOI 10.1073/pnas.2518204122.
S03 2024	Systematic review of 58 studies using SAMR	ITS often augment existing practice through personalisation and feedback.	Classifying technology integration does not estimate a causal effect.	Describe the learning and assessment cycle instead of claiming that AI itself is novel.
S04 2024	Mixed methods: 5 interviews and 616 written explanations	Carefully designed distractors can identify a specific algebraic misconception.	The validity of one test does not automatically transfer to fractions or another language.	Develop skill-specific distractors and structured criteria, cross-review them within the team and evaluate diagnoses on new items. Internal review does not establish external validity.
S05 2024	Knowledge tracing model evaluated on three public datasets	Skill–item relationships support next-answer prediction with reduced computational cost.	Predicting answers does not establish understanding; the model needs interaction history.	Store item–skill mappings. Start with a transparent rule and defer ML until sufficient data exist.

Code and year	Method	Finding	Limitation	Design response
S06 2024	Two randomised controlled trials with 404 students in grades 5–6	Additional mindfulness mechanics produced no significant between-condition differences in learning gains.	Specific mechanics were studied, not gamification as a whole; the mindfulness induction was unsuccessful.	Evaluate learning separately from game appeal.
S07 2025	Logs and observations in grade 5 and grade 6 classes of 30 and 22 students	LMS records capture open collaborative activity poorly; learning context affects interpretation.	Two classes; activity outside the system is not captured. Published online on 31 December 2024, in a 2025 volume.	Label time as activity within the platform and show the amount of diagnostic evidence.
S08 2024	Systematic review of 31 studies	Engagement and motivation often improve; learning outcomes are mixed.	Different subjects and mechanics; transfer to a specific product requires evaluation.	Treat XP and achievements as motivational features; assess mastery with separate items.
S09 2024	Qualitative meta-synthesis of 47 publications	School analytics can support personalisation but introduces interpretation, privacy and equity risks.	A review of opportunities does not establish the effectiveness of every dashboard.	Minimise data, separate roles and test report comprehension.
S10 2024	Study of teachers' use of visualisations	Visualisations helped identify difficulties and plan feedback; teachers rechecked uncertain data.	Teachers and the topic of functions; findings do not prove the usefulness of a parent interface.	Show attempts and supporting evidence rather than a single opaque rating.
S11 2025	PRISMA review of 21 studies	Dashboards use prediction and support self-regulation; causal and field evaluations are often missing.	Small samples and weak links between prediction and validated intervention.	Evaluate whether parents can choose the correct next step from the report.

Code and year	Method	Finding	Limitation	Design response
S12 2025	Two preschool cohorts using apps and parent information	The intervention showed larger numeracy gains; improvement in the home numeracy environment was not found.	Preschool participants; the effect cannot be attributed to the parent dashboard alone.	Give families specific recommendations and test the dashboard in the target age group.

Additional Official Evidence

[O01] OECD PISA 2022 provides context on mathematical literacy, with limitations concerning participants' age and the year of the study.

[O02] UNESCO Guidance for Generative AI in Education and Research supports age-appropriate and pedagogical controls and minimisation of transmitted data. These are recommendations, not a legal opinion. The selected AI provider's rules and local requirements must be checked separately before children participate.

References

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[S08] Romero-Rodríguez, J.-M., Martínez-Menéndez, A., Alonso-García, S., & Victoria-Maldonado, J.-J. (2024). The reality of the gamification methodology in Primary Education: A systematic review. International Journal of Educational Research, 128, 102481. DOI: https://doi.org/10.1016/j.ijer.2024.102481.

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